Class 10 - Solutions

April 14, 2026

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Statistical Learning

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Machine Learning and Econometrics

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ECO 4971/6971

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Class 10 — Model Selection (Ridge, Lasso): Exercise Solutions

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These exercises use the `Credit.csv` dataset.

1 Exercise 1 — Best Subset Selection

Q1. Which single variable does the best job predicting `Balance`?

Answer.

Best-subset selection with one predictor picks the variable with the highest R^2 (equivalently, lowest RSS) when regressed individually on `Balance`. In `Credit.csv`, the variable with the strongest simple linear relationship with credit card balance is `Rating` (credit rating), which has a correlation of roughly 0.86 with `Balance`.

`Income` is the second strongest individual predictor. The remaining variables (Limit, Age, Cards, Education, etc.) are weaker individually.

Q2. How many possible subsets are there if you have k predictors?

Answer.

For each predictor we make a binary choice: include it or not. With k predictors there are 2^k total subsets (including the null model with no predictors). Excluding the null model, there are $2^k - 1$ non-empty subsets.

Predictors (k)	Non-empty subsets ($2^k - 1$)
1	1
2	3
3	7
4	15
5	31

The number grows exponentially. The `Credit` dataset has 10 predictors, giving $2^{10} - 1 = 1{,}023$ subsets — feasible, but with $p = 30$ predictors we already have over 10^9 subsets, making exhaustive search computationally infeasible. This motivates **stepwise** and **shrinkage** methods.

2 Exercise 2 — Ridge Regression

Q3. Why do we call Ridge regression a "shrinkage" estimator?

Answer.

Ridge regression minimises a penalised criterion:

$$\text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|\boldsymbol{\beta}\|_2^2$$

The ℓ_2 penalty $\lambda \sum_j \beta_j^2$ penalises large coefficients. As λ increases, the optimal $\hat{\boldsymbol{\beta}}^{\text{Ridge}}$ is pushed toward zero — the coefficients are **shrunk** relative to the OLS estimates $\hat{\boldsymbol{\beta}}^{\text{OLS}}$.

The closed-form solution makes this explicit:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

When $\lambda = 0$ this reduces to OLS; as $\lambda \to \infty$ all coefficients shrink to 0. The name "shrinkage" refers specifically to this pulling of estimates toward zero.

Q4. Suppose we predict farm harvest with:

$$\text{TotalHarvest} = \beta_0 + \beta_1 \,\text{AverageTemperature} + \beta_2 \,\text{AverageRainfall} + u$$

(a) If I run OLS with temperature in °C and then in °K, what changes?

Answer (a).

Kelvin = Celsius + 273.15, a **linear (affine) transformation**. OLS is **equivariant** under linear transformations of the predictors: the coefficient on Temperature simply rescales to keep the fitted values identical.

- $\hat{\beta}_1^{(K)} = \hat{\beta}_1^{(C)}$ — since the *slope* does not change when we add a constant (the constant is absorbed into the intercept).
- The **intercept changes**: $\hat{\beta}_0^{(K)} = \hat{\beta}_0^{(C)} - 273.15\,\hat{\beta}_1$.
- All other quantities — fitted values, residuals, R^2, p-values — are **identical**.

(For a unit rescaling like °C → °F the slope would change by the conversion factor, but the fitted values and R^2 remain identical.)

(b) If I run Ridge regression with temperature in °C and then °K, what changes?

Answer (b).

Ridge is **not** equivariant under rescaling. The ℓ_2 penalty $\lambda(\beta_1^2 + \beta_2^2)$ penalises the raw coefficient values. If we change temperature from °C to °K (adding 273.15), the scale of X_1 changes, so the ridge penalty treats the same physical information differently depending on units.

Concretely, the same λ will impose a *different* amount of shrinkage on $\hat{\beta}_1$ depending on whether temperature is in °C or °K. The fitted values and test MSE will differ.

This is why we always standardise predictors before applying Ridge (or Lasso): standardisation ensures all predictors enter the penalty on equal footing, making the shrinkage unit-free.

3 Exercise 3 — Lasso

Q5. What is the difference between Lasso and Ridge regression?

Answer.

Both methods add a penalty to OLS to shrink coefficients, but they differ in the **type of penalty**:

	Ridge	**Lasso**
Penalty	$\lambda \sum_j \beta_j^2$ (ℓ_2 norm)	$\lambda \sum_j \lvert\beta_j\rvert$ (ℓ_1 norm)
Coefficients	Shrunk toward zero, never exactly zero	Can be set **exactly to zero**
Variable selection	No (all variables retained)	**Yes** (sparse solutions)
Geometry	Circular constraint region	Diamond constraint region — corners at axes produce zeros
Best when	Many small effects (dense signal)	Few large effects, many irrelevant predictors (sparse signal)

The key practical difference: **Lasso performs automatic variable selection** by setting irrelevant coefficients to exactly zero, producing an interpretable sparse model. Ridge keeps all variables but dampens their influence.

Q6. How would you decide among Lasso, Ridge, and OLS in practice?

Answer.

Use **cross-validation** to compare prediction error across the three approaches (and across values of λ for Ridge and Lasso). The practical decision logic:

1. **When to prefer OLS:** Few predictors relative to observations, no reason to believe many are irrelevant. OLS is unbiased and interpretable; regularisation would only add unnecessary bias.
2. **When to prefer Ridge:** Many predictors that may all contribute a little (dense signal), or multicollinearity among predictors. Ridge stabilises estimates by shrinking correlated coefficients together.
3. **When to prefer Lasso:** Many predictors but only a few are truly relevant (sparse signal). Lasso's variable selection yields an interpretable model and can improve prediction when many predictors are noise.
4. **Elastic Net** (a blend of Ridge and Lasso penalties) is useful when predictors are correlated and sparsity is still desired — Lasso tends to arbitrarily pick one among highly correlated predictors, while Elastic Net can retain groups.

In all cases, **tune λ via cross-validation** on a training set and report final performance on a held-out test set.